A hybrid neural simulation-based inference method for LHC analyses

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(Dated: July 16, 2026)

Neural simulation-based inference based on classifier estimates of density ratios has made unbinned, high-dimensional likelihood-ratio inference practical for analyses with many physics processes and nuisance parameters. Its ratio-only formulation, however, does not provide an explicit normalized event density or a direct generative model. Expected-data calculations and frequentist pseudo-experiments must therefore be represented by large weighted simulated samples, which can dominate the cost of profiling, uncertainty decomposition, and Neyman constructions. Normalizing flows provide normalized densities and efficient sampling, but small absolute-density errors can be consequential when a small signal is inferred in a much larger background.

We propose *hybrid neural simulation-based inference*. A normalizing flow is trained on a balanced, physically representative reference distribution. Events sampled from that learned reference—rather than the finite simulation used to train it—form the denominator sample for classifier estimates of each process-to-reference density ratio. A process density is represented by the tractable reference-flow density multiplied by its learned ratio. This division of labor preserves the discriminative precision of density-ratio estimation while making every component of the event intensity tractable and sampleable. The flow supplies arbitrarily large reference quadratures for weighted, unbinned Asimov datasets and for pseudo-experiment generation. We further introduce a neural importance sampler optimized for the expected likelihood-ratio scan. In a five-dimensional small-signal demonstration, the finite weighted Asimov sample fits exactly at its generating value, predicts the width and discovery-statistic distribution observed in pseudo-experiments, and neural importance sampling reduces the variance of the discovery statistic by a factor of 2.69 at fixed sample size. A flow thus makes unbinned Asimov inference cheap to construct, while neural importance sampling makes the resulting quadrature statistically efficient.

I. INTRODUCTION

Likelihood-based measurements at the Large Hadron Collider (LHC) conventionally compress reconstructed events into one or a few binned observables. This construction is robust and computationally convenient, but the compression and binning can discard information when the likelihood depends nonlinearly on parameters or when no low-dimensional summary is sufficient over the full parameter space. Neural simulation-based inference (NSBI) instead uses high-dimensional simulated events to estimate likelihoods or likelihood ratios directly in reconstructed feature space [1–3].

The ATLAS Collaboration developed a practical classifier-based NSBI formalism for LHC parameter estimation [4]. It was deployed in the measurement of off-shell Higgs-boson production in the $H^* \to ZZ \to 4\ell$ final state [5]. The formalism uses neural estimates of density ratios and an analytic factorization of the dependence on parameters of interest (POIs) and nuisance parameters. It can therefore describe dozens of POIs and hundreds of nuisance parameters without densely sampling that complete parameter space during neural-network training. We refer to this construction as *parametrized NSBI*; it has also been described as a factorized or semi-parametric approach [6].

Density ratios are sufficient for inference based on a profile likelihood ratio. They are also often easier to estimate to the required precision than absolute densities. A reference sample can be designed to resemble the relevant physics, sharing resonant structure, tails, and detector effects with the target samples. The resulting ratios vary more mildly than either absolute density, and they can be learned discriminatively with a binary cross-entropy objective. This numerical advantage was central to the precision achieved in the ATLAS studies.

A ratio-only representation nevertheless has an important operational limitation: it is not itself a generative density model. An unbinned Asimov dataset must be approximated by a large weighted Monte Carlo (MC) sample, and pseudo-experiments must be constructed by resampling such a sample. The former makes repeated profiling and impact calculations costly; the latter requires an MC pool much larger than a typical pseudo-experiment [4]. These difficulties do not invalidate ratio-based inference, but they make some standard LHC statistical workflows expensive.

Normalizing flows address the complementary problem. They provide both a normalized, tractable density and efficient event generation [7–10]. The difficulty is precision: two separately trained process flows can each pass many marginal checks while their residual, unrelated absolute-density errors bias the process-density ratio that controls a small-signal measurement. Recent work has developed factorized parameter-dependent flows [11], but the required absolute-density accuracy remains a central question when the signal is orders of magnitude smaller than the background.

This paper combines the two representations. A flow is used to build a normalized, sampleable reference density, while classifiers learn the precision-critical ratios of each process to that exact learned reference. The reference flow is a proposal distribution, not an uncorrected model of every process. If the ratio estimators are exact, local reference-flow error cancels algebraically. What the flow must provide is common support and efficient coverage. The resulting *hybrid NSBI* construction supplies a tractable likelihood, direct weighted or resampled event generation, and a deterministic unbinned Asimov measure.

The same formulation also exposes the Asimov calculation as numerical integration. Direct samples from the reference flow make this integration inexpensive and remove the resolution limit of a finite MC pool, but ordinary Monte Carlo still converges as $M^{-1/2}$. We therefore train a second flow as a neural importance proposal for the complete expected test-statistic scan. This distinction is useful in practice: the reference flow makes an Asimov dataset *cheap*, because more independent quadrature points can be produced on demand; the importance proposal makes it *efficient*, because fewer such points are required for a target precision.

The remainder of this paper is organized as follows. Section II reviews parametrized NSBI and its practical limitations. Section III introduces the hybrid density and sampling construction. Section IV derives the weighted unbinned Asimov likelihood and its finite-sample uncertainty. Section V develops neural importance sampling for expected likelihood scans. Section VI reports the five-dimensional proof-of-concept experiments, followed by discussion and conclusions.

II. PARAMETRIZED NEURAL SIMULATION-BASED INFERENCE

A. Extended unbinned likelihood

Let $\boldsymbol{x} \in \mathbb{R}^d$ denote the reconstructed observables for an event and let j index physics components. For parameters $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\alpha})$, including POIs $\boldsymbol{\mu}$ and nuisance parameters $\boldsymbol{\alpha}$, define the differential event intensity

$$\nu(\boldsymbol{x} \mid \boldsymbol{\theta}) = \sum_j \lambda_j(\boldsymbol{\theta}) p_j(\boldsymbol{x} \mid \boldsymbol{\theta}), \tag{1}$$

where p_j is normalized to unity and λ_j is the expected yield. Its integral is

$$\Lambda(\boldsymbol{\theta}) = \int \nu(\boldsymbol{x} \mid \boldsymbol{\theta})\, \mathrm{d}\boldsymbol{x} = \sum_j \lambda_j(\boldsymbol{\theta}). \tag{2}$$

Up to factors independent of $\boldsymbol{\theta}$, the extended unbinned likelihood for n observed events is

$$\mathcal{L}(\boldsymbol{\theta}) = e^{-\Lambda(\boldsymbol{\theta})} \prod_{i=1}^{n} \nu(\boldsymbol{x}_i \mid \boldsymbol{\theta}) \prod_k \pi_k(a_k \mid \alpha_k), \tag{3}$$

where π_k are constraint terms for auxiliary measurements a_k. The profile-likelihood-ratio statistic for a POI value $\boldsymbol{\mu}$ is

$$t_{\boldsymbol{\mu}} = -2 \log \frac{\mathcal{L}(\boldsymbol{\mu}, \widehat{\widehat{\boldsymbol{\alpha}}}_{\boldsymbol{\mu}})}{\mathcal{L}(\widehat{\mu}, \widehat{\boldsymbol{\alpha}})}. \tag{4}$$

B. Classifiers as density-ratio estimators

Consider a classifier trained with equal total class weight to distinguish events from densities p_1 and p_0. At the population minimum of the binary cross-entropy loss, its score is

$$s^*(\boldsymbol{x}) = \frac{p_1(\boldsymbol{x})}{p_1(\boldsymbol{x}) + p_0(\boldsymbol{x})}. \tag{5}$$

The density ratio follows from the odds [1],

$$r(\boldsymbol{x}) = \frac{p_1(\boldsymbol{x})}{p_0(\boldsymbol{x})} = \frac{s^*(\boldsymbol{x})}{1 - s^*(\boldsymbol{x})}. \tag{6}$$

Balanced class weights are important: otherwise the class-prior odds must be included explicitly. In practice, ensembles, calibration tests, and reweighting closure tests are used to control finite-sample, optimization, and model-capacity effects [4].

Choose a fixed reference density $p_{\mathrm{ref}}(\boldsymbol{x})$ whose support covers all relevant components, and define

$$r_j^{\mathrm{ref}}(\boldsymbol{x} \mid \boldsymbol{\theta}) = \frac{p_j(\boldsymbol{x} \mid \boldsymbol{\theta})}{p_{\mathrm{ref}}(\boldsymbol{x})}. \tag{7}$$

The intensity may then be written

$$\nu(\boldsymbol{x} \mid \boldsymbol{\theta}) = p_{\mathrm{ref}}(\boldsymbol{x}) \sum_j \lambda_j(\boldsymbol{\theta}) r_j^{\mathrm{ref}}(\boldsymbol{x} \mid \boldsymbol{\theta}). \tag{8}$$

Because the reference is parameter independent, $\sum_i \log p_{\mathrm{ref}}(\boldsymbol{x}_i)$ cancels from every likelihood ratio. Hence the reference need not have a tractable density for the inference in Eq. (4); only the component-to-reference ratios are required.

C. Factorized parameter dependence

Directly training a single conditional estimator throughout a space with many POIs and nuisance parameters generally requires dense simulated coverage of that space. Parametrized NSBI instead exploits the analytic structure of an LHC likelihood. A useful schematic form is

$$\frac{p(\boldsymbol{x} \mid \boldsymbol{\mu}, \boldsymbol{\alpha})}{p_{\mathrm{ref}}(\boldsymbol{x})} = \frac{1}{\Lambda(\boldsymbol{\mu}, \boldsymbol{\alpha})} \sum_j f_j(\boldsymbol{\mu}) \lambda_j r_j^0(\boldsymbol{x}) \times \prod_k G_{jk}(\alpha_k) g_{jk}(\boldsymbol{x}, \alpha_k). \tag{9}$$

Here f_j describes known POI dependence, G_{jk} is a yield interpolation, $r_j^0 = p_j^0/p_{\rm ref}$ is a nominal process-to-reference ratio, and $g_{jk} = p_j(\boldsymbol{x} \mid \alpha_k)/p_j^0(\boldsymbol{x})$ describes a systematic shape variation. The individual ratios can be trained on nominal and varied samples; their parameter dependence is supplied by low-dimensional analytic interpolation. This is a simplification of fully conditional, simulator-assisted methods [2, 3], but it matches how systematic variations are made available in full LHC analyses and scales to large nuisance-parameter models [4, 6].

D. Operational limitations of a ratio-only model

The cancellation of $p_{\rm ref}$ in Eq. (8) is an advantage for parameter estimation, but it also means that the construction does not specify a normalized density that can be evaluated or sampled. Two consequences are particularly relevant.

First, an Asimov dataset is meant to return the generating parameter without statistical fluctuation and to encode median expected sensitivity [12]. In a binned likelihood, the expected count is placed in every bin. In an unbinned ratio-only analysis, one instead uses a large weighted MC sample as numerical quadrature for the expected event measure. The approximation improves as this sample grows, but evaluating hundreds of neural networks on millions of events and repeating the calculation for profiles or impact decompositions can be expensive. The ATLAS implementation reports likelihood maximizations ranging from minutes and gigabytes to hours and hundreds of gigabytes as the selected sample grows from thousands to millions of events [4].

Second, a Neyman construction requires samples from the model at each tested parameter point. A ratio alone does not provide a direct sampler. A valid workaround is to draw with replacement from a much larger positive-weight reference pool, with probabilities proportional to event weights or learned ratios. The ATLAS procedure assigns each reference event a Poisson-distributed integer multiplicity with mean equal to its Asimov weight [4]. This is statistically sound for the discrete MC approximation but requires a pool much larger than an individual pseudo-experiment, and the pool remains the resolution limit of the generator.

These are computational and representational limitations, not failures of density-ratio inference. They motivate adding a generative reference model without surrendering the ratio precision that made parametrized NSBI practical.

III. HYBRID NEURAL SIMULATION-BASED INFERENCE

A. A tractable reference density

Let $q_\phi(\boldsymbol{x})$ be a normalized normalizing-flow density trained on a physically representative reference sample. For a signal-plus-background analysis, a useful target after the event selection is

$$p_{\rm mix}(\boldsymbol{x}) = \tfrac{1}{2}p_S(\boldsymbol{x}) + \tfrac{1}{2}p_B(\boldsymbol{x}). \tag{10}$$

The equal mixture is deliberate. A mixture with physical process yields would be almost pure background when $S/B \ll 1$ and would allocate little flow capacity to signal-rich regions. Equal component normalization instead provides comparable coverage of both phase spaces.

After training, a dedicated sample is drawn from q_ϕ. This *flow-generated* sample, rather than the MC sample used to fit the flow, supplies the denominator class for each classifier. The corresponding process ratio and hybrid density are

$$r_j(\boldsymbol{x}) = \frac{p_j(\boldsymbol{x})}{q_\phi(\boldsymbol{x})}, \qquad \widehat{p}_j(\boldsymbol{x}) = q_\phi(\boldsymbol{x})\widehat{r}_j(\boldsymbol{x}). \tag{11}$$

This ordering is essential. Even if $q_\phi \neq p_{\rm mix}$, an exact classifier trained against events from q_ϕ learns the correction p_j/q_ϕ, so their product recovers p_j. The flow is not required to be an independently precise model of every physics component. It must be tractable, efficiently sampleable, and nonzero wherever a target component is nonzero.

The last condition is not merely formal. If p_j is appreciable where q_ϕ is extremely small, the learned ratio develops long tails, classifier training becomes difficult, and any subsequent importance sampler has low effective sample size. Balanced mixture design, broad flow tails, and ratio-tail diagnostics are therefore part of the method.

B. Likelihood and empirical normalization

Define

$$H_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sum_j \lambda_j(\boldsymbol{\theta})\widetilde{r}_j(\boldsymbol{x} \mid \boldsymbol{\theta}). \tag{12}$$

The hybrid event intensity is

$$\widehat{\nu}(\boldsymbol{x} \mid \boldsymbol{\theta}) = q_\phi(\boldsymbol{x})H_{\boldsymbol{\theta}}(\boldsymbol{x}). \tag{13}$$

For a fixed reference flow, $\sum_i \log q_\phi(\boldsymbol{x}_i)$ is independent of $\boldsymbol{\theta}$ and cancels from profile likelihood ratios. Thus ordinary inference retains the numerical structure of parametrized NSBI,

$$\begin{aligned} -2\log\mathcal{L}(\boldsymbol{\theta}) = 2\Lambda(\boldsymbol{\theta}) &- 2\sum_i \log H_{\boldsymbol{\theta}}(\boldsymbol{x}_i) \\ &- 2\sum_k \log \pi_k(a_k \mid \alpha_k) + \text{constant}, \end{aligned} \tag{14}$$

The explicit factor q_ϕ becomes relevant when an absolute density, integral, or sampler is needed.

An exact component ratio satisfies $\mathbb{E}_{q_\phi}[r_j] = 1$. Let $\{\boldsymbol{x}_m, \omega_m\}_{m=1}^M$ be a quadrature for expectation under q_ϕ, with $\omega_m \geq 0$ and $\sum_m \omega_m = 1$. For direct independent samples from q_ϕ, $\omega_m = 1/M$. We use the empirical normalization

$$\widetilde{r}_j(\boldsymbol{x}) = \frac{\widehat{r}_j(\boldsymbol{x})}{\sum_{m=1}^M \omega_m \widehat{r}_j(\boldsymbol{x}_m)}, \qquad \sum_m \omega_m \widetilde{r}_j(\boldsymbol{x}_m) = 1. \tag{15}$$

This operation removes a finite-sample normalization offset but does not repair local classifier bias. Calibration, ensemble stability, multidimensional reweighting, and inference-level closure remain necessary. The same factorized POI and nuisance-parameter model as in Eq. (9) can be placed inside $H_{\boldsymbol{\theta}}$, so the hybrid reference is compatible with analytic interpolation, conditional estimators, or nonparametric parameter dependence [11, 13].

C. Sampling and Neyman construction

A fixed direct-flow sample represents component j with probabilities

$$w_m^{(j)} = \frac{\widetilde{r}_j(\boldsymbol{x}_m)}{M}, \qquad \sum_{m=1}^M w_m^{(j)} = 1. \tag{16}$$

Weighted expectations then approximate integrals over p_j. An unweighted component sample can be obtained by resampling indices with probabilities $w_m^{(j)}$; a fresh pool can be generated from the flow whenever duplicate rates become excessive. Rejection sampling is another option when a reliable upper bound on the ratio is available.

For a pseudo-experiment at $\boldsymbol{\theta}$, component counts are drawn independently,

$$n_j \sim \mathrm{Pois}(\lambda_j(\boldsymbol{\theta}))\,, \tag{17}$$

followed by n_j draws from the corresponding component probabilities. Concatenating the components gives an unweighted pseudo-dataset from the empirical hybrid model. Repeated likelihood fits provide the sampling distribution of $t_{\boldsymbol{\mu}}$ for a Neyman construction [14].

The component effective sample size,

$$M_{\text{eff}}^{(j)} = \frac{\left(\sum_m \widetilde{r}_j\right)^2}{\sum_m \widetilde{r}_j^2}, \tag{18}$$

and the upper ratio quantiles diagnose proposal coverage. In contrast with a finite MC reference, the flow is not the resolution limit: additional independent proposal points can be generated on demand.

IV. WEIGHTED UNBINNED ASIMOV INFERENCE

A. Expected event measure and extended likelihood

An unbinned Asimov dataset is most naturally an expected event *measure*, rather than a special random list of unweighted events. At a generating point $\boldsymbol{\theta}_A$,

$$\mathrm{d}N_A(\boldsymbol{x}) = \widehat{\nu}(\boldsymbol{x} \mid \boldsymbol{\theta}_A)\,\mathrm{d}\boldsymbol{x} = q_\phi(\boldsymbol{x}) H_A(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x}, \tag{19}$$

where $H_A \equiv H_{\boldsymbol{\theta}_A}$. The quadrature $\{\boldsymbol{x}_m, \omega_m\}$ represents this measure with event weights

$$w_m^A = \omega_m H_A(\boldsymbol{x}_m). \tag{20}$$

For any integrable function f,

$$\int f(\boldsymbol{x})\,\mathrm{d}N_A(\boldsymbol{x}) \simeq \sum_m w_m^A f(\boldsymbol{x}_m). \tag{21}$$

Because the component ratios are normalized on the same quadrature,

$$\sum_m w_m^A = \sum_j \lambda_j(\boldsymbol{\theta}_A) = \Lambda(\boldsymbol{\theta}_A), \tag{22}$$

so the Asimov sample contains exactly the expected number of weighted events. This is how the extended part is represented.

With auxiliary data fixed to their expectation, the finite weighted log likelihood is

$$\begin{aligned} \log \mathcal{L}_A(\boldsymbol{\theta}) = &-\Lambda(\boldsymbol{\theta}) + \sum_m w_m^A \log \widehat{\nu}(\boldsymbol{x}_m \mid \boldsymbol{\theta}) \\ &+ \log \pi(\boldsymbol{a}_A \mid \boldsymbol{\alpha}), \end{aligned} \tag{23}$$

which is the direct unbinned analogue of filling each bin with its expected count.

B. Expected test statistic and quadrature uncertainty

The continuous expected log likelihood is, up to parameter-independent terms,

$$\ell_A(\boldsymbol{\theta}) = -\Lambda(\boldsymbol{\theta}) + \int \nu_A(\boldsymbol{x}) \log \nu(\boldsymbol{x} \mid \boldsymbol{\theta})\,\mathrm{d}\boldsymbol{x}. \tag{24}$$

Assuming it is maximized at $\boldsymbol{\theta}_A$, direct subtraction gives

$$\begin{aligned} t_A(\boldsymbol{\theta}) &= -2\left[\ell_A(\boldsymbol{\theta}) - \ell_A(\boldsymbol{\theta}_A)\right] \\ &= 2\left\{\Lambda(\boldsymbol{\theta}) - \Lambda(\boldsymbol{\theta}_A) + \int \nu_A(\boldsymbol{x}) \log \frac{\nu_A(\boldsymbol{x})}{\nu(\boldsymbol{x} \mid \boldsymbol{\theta})}\,\mathrm{d}\boldsymbol{x}\right\} \\ &= 2\left\{\Lambda(\boldsymbol{\theta}) - \Lambda(\boldsymbol{\theta}_A) + \mathbb{E}_{q_\phi}\left[H_A(\boldsymbol{x}) \log \frac{H_A(\boldsymbol{x})}{H_{\boldsymbol{\theta}}(\boldsymbol{x})}\right]\right\}. \end{aligned} \tag{25}$$

The common factor q_ϕ cancels inside the logarithm. The first two terms are the extended-rate contribution; the expectation is the shape-and-composition contribution. The formula also exposes the expected scan as an ordinary expectation under a density that can be sampled without returning to detector simulation.

For direct independent points $\boldsymbol{x}_m \sim q_\phi$, define

$$Y_{\boldsymbol{\theta}}(\boldsymbol{x}) = H_A(\boldsymbol{x}) \log \frac{H_A(\boldsymbol{x})}{H_{\boldsymbol{\theta}}(\boldsymbol{x})}. \tag{26}$$

The Monte Carlo estimator is

$$\widehat{t}_A(\boldsymbol{\theta}) = 2\left[\Lambda(\boldsymbol{\theta}) - \Lambda(\boldsymbol{\theta}_A) + \frac{1}{M}\sum_{m=1}^{M} Y_{\boldsymbol{\theta}}(\boldsymbol{x}_m)\right]. \tag{27}$$

Only the sample mean fluctuates. Hence

$$\mathrm{Var}\big[\widehat{t}_A(\boldsymbol{\theta})\big] = \frac{4}{M}\,\mathrm{Var}_{q_\phi}[Y_{\boldsymbol{\theta}}], \qquad \mathrm{SE}[\widehat{t}_A(\boldsymbol{\theta})] \simeq \frac{2 s_Y(\boldsymbol{\theta})}{\sqrt{M}}, \tag{28}$$

where

$$s_Y^2(\boldsymbol{\theta}) = \frac{1}{M-1}\sum_m \left[Y_{\boldsymbol{\theta}}(\boldsymbol{x}_m) - \overline{Y}_{\boldsymbol{\theta}}\right]^2. \tag{29}$$

Thus s_Y is the ordinary sample standard deviation, under the reference density, of the event-level contribution to the expected log-likelihood ratio. It depends on the tested parameter. Points on a scan are strongly correlated because they use the same events; independent repeats or block estimates are preferable for uncertainty on the complete curve. If the same finite sample normalizes the learned ratios, the summands are weakly coupled and Eq. (28) is asymptotic; blocks or repeated quadratures capture this additional dependence.

C. Exact stationarity for yield parameters

For the one-parameter model used below,

$$H_\mu(\boldsymbol{x}) = \mu\lambda_S \widetilde{r}_S(\boldsymbol{x}) + \lambda_B \widetilde{r}_B(\boldsymbol{x}). \tag{30}$$

Differentiating Eq. (23) at μ_A gives

$$-2\left.\frac{\partial \log \mathcal{L}_A}{\partial \mu}\right|_{\mu_A} = 2\lambda_S\left[1 - \sum_m \omega_m \widetilde{r}_S(\boldsymbol{x}_m)\right] = 0. \tag{31}$$

The equality is exact because the signal ratio was normalized on this quadrature. The second derivative is nonnegative, so the finite weighted Asimov likelihood has its minimum at the generating value up to optimizer precision. The same argument applies to several linearly entering component normalizations: differentiation with respect to each yield multiplier selects the corresponding normalized ratio. Shape-changing nuisance parameters retain exact expected-score closure in the continuous limit, but a finite quadrature has residual integration error unless its nuisance scores are also balanced. Auxiliary measurements must, as usual, be fixed to their expectation.

D. Expected uncertainty and toy distributions

Following Cowan *et al.* [12], an Asimov sample generated at $\mu_A > 0$ predicts the median discovery statistic and local uncertainty,

$$q_{0,A} = -2\log\frac{\mathcal{L}_A(0)}{\mathcal{L}_A(\widehat{\mu}_A)}, \quad \sigma_A \simeq \frac{\mu_A}{\sqrt{q_{0,A}}}, \quad Z_A = \sqrt{q_{0,A}}. \tag{32}$$

For an unconstrained two-sided statistic, t_0 approaches a noncentral χ^2 distribution with one degree of freedom and noncentrality $q_{0,A}$. With the physical constraint $\widehat{\mu} \geq 0$, the discovery statistic is instead represented asymptotically by

$$q_0 = [\max(0, Z)]^2, \qquad Z \sim \mathcal{N}(\sqrt{q_{0,A}}, 1), \tag{33}$$

including a point mass $\Phi(-\sqrt{q_{0,A}})$ at zero.

V. NEURAL IMPORTANCE SAMPLING FOR EFFICIENT ASIMOV SCANS

A. Variance-optimal proposal

Direct sampling from q_ϕ solves the availability problem, but it does not guarantee that a small quadrature resolves the regions that dominate Eq. (25). Consider K scan points $\{\boldsymbol{\theta}_k\}$ and the integrals

$$I_k = \mathbb{E}_{q_\phi}[Y_k(\boldsymbol{x})], \qquad Y_k \equiv Y_{\boldsymbol{\theta}_k}. \tag{34}$$

For samples from another normalized proposal $g(\boldsymbol{x})$ with common support,

$$\widehat{I}_k = \frac{1}{M}\sum_{m=1}^{M}\frac{q_\phi(\boldsymbol{x}_m)}{g(\boldsymbol{x}_m)} Y_k(\boldsymbol{x}_m), \qquad \boldsymbol{x}_m \sim g. \tag{35}$$

The second moment controlling its variance is

$$\mathbb{E}_g\left[\left(\frac{q_\phi}{g}Y_k\right)^2\right] = \int \frac{q_\phi(\boldsymbol{x})^2 Y_k(\boldsymbol{x})^2}{g(\boldsymbol{x})}\,\mathrm{d}\boldsymbol{x}. \tag{36}$$

For one integral, variational minimization under $\int g = 1$ gives $g^* \propto q_\phi |Y_k|$. For an entire scan, minimizing a weighted sum of the variances gives

$$g^*(\boldsymbol{x}) \propto q_\phi(\boldsymbol{x}) A(\boldsymbol{x}), \qquad A(\boldsymbol{x}) = \left[\sum_{k=1}^{K} a_k Y_k(\boldsymbol{x})^2\right]^{1/2}, \tag{37}$$

where a_k specifies the relative precision desired at each scan point. The proposal targets events important to at least one point, rather than concentrating only on $q_{0,A}$. Overall scales in Y_k may be removed for numerical stability without changing the normalized target.

The density in Eq. (37) is known pointwise up to a normalization constant: q_ϕ is tractable and A is computed from the learned ratios. A second normalizing flow g_ψ can therefore be trained on a pilot reference sample resampled in proportion to A. This is neural importance sampling [15]; related uses in collider inference include the SPINUP unfolding construction [16]. Residual proposal error does not bias the raw estimator in Eq. (35), because the exact weight q_ϕ/g_ψ corrects it.

B. Defensive sampling and finite quadrature

An approximate proposal should be protected against missed support. We use the defensive mixture

$$g_\epsilon(\boldsymbol{x}) = (1-\epsilon)g_\psi(\boldsymbol{x}) + \epsilon q_\phi(\boldsymbol{x}), \tag{38}$$

which guarantees $q_\phi/g_\epsilon \le 1/\epsilon$. For nodes $\boldsymbol{x}_m \sim g_\epsilon$, either the unbiased weights $(q_\phi/g_\epsilon)/M$ or the self-normalized quadrature weights

$$\omega_m = \frac{\rho_m}{\sum_\ell \rho_\ell}, \qquad \rho_m = \frac{q_\phi(\boldsymbol{x}_m)}{g_\epsilon(\boldsymbol{x}_m)} \tag{39}$$

can be used. The latter have an $O(M^{-1})$ ratio-estimator bias but are often more stable. Normalizing every process ratio with the same ω_m according to Eq. (15) preserves the exact finite-sample score property in Eq. (31). The relevant weight diagnostic is

$$M_{\text{eff}} = \frac{(\sum_m \rho_m)^2}{\sum_m \rho_m^2}. \tag{40}$$

High effective sample size alone does not demonstrate variance reduction, since a proposal equal to q_ϕ has $M_{\text{eff}} = M$. It must be combined with target closure and repeated equal-cost estimates of the statistic of interest.

C. A practical recipe for an LHC analysis

For a physicist implementing the method, the following sequence separates model validation from integration optimization.

1. *Freeze the likelihood model.* Validate q_ϕ and every process ratio using independent samples. Choose the Asimov point $\boldsymbol{\theta}_A$, the POI range, and the nuisance-parameter treatment before optimizing the quadrature.

2. *Build a pilot target.* Draw a moderately large pilot sample from q_ϕ, evaluate all ratios, and calculate Y_k on a grid of POI values. Form A in Eq. (37); clip an extreme upper tail and impose a small positive floor so that the training target remains numerically stable.

3. *Train the importance flow.* Resample pilot points with probabilities proportional to A and fit g_ψ by maximum likelihood. This step changes only the quadrature, not the physics likelihood.

4. *Sample defensively.* Generate nodes from g_ϵ with a nonzero reference fraction, evaluate $\log q_\phi - \log g_\epsilon$, and construct ω_m. Normalize all process ratios on this exact weighted quadrature.

5. *Validate the integration.* Check closure of reweighted feature distributions, the weight tails and M_{eff}, the score at $\boldsymbol{\theta}_A$, and repeated independent likelihood scans. Quote a block or repetition uncertainty on $q_{0,A}$ and on representative scan points.

6. *Demonstrate efficiency.* Compare with direct q_ϕ sampling at equal numbers of ratio evaluations and, for production, at equal wall time. The proposal-training cost is worthwhile when the same quadrature is reused for scans, profiling, systematic impacts, or combinations.

This leads to a concise division of labor. The reference flow makes an Asimov calculation cheap to instantiate and arbitrarily extensible; neural importance sampling makes it efficient at fixed quadrature size. The two statements are distinct. A finite MC reference can also define a correct Asimov quadrature, but it cannot produce new independent phase-space points once generated. Conversely, a poor neural proposal remains correct after exact reweighting but may offer no efficiency gain.

VI. FIVE-DIMENSIONAL DEMONSTRATION

We implement the hybrid construction and its neural importance extension in Exercises 5 and 6 of the ML4HEP 2026 conference tutorial [6, 17, 18]. Analytic reconstructed densities are available throughout, so flow, ratio, likelihood, Asimov, and pseudo-experiment closure can be tested independently.

A. Samples and preselection

The toy measurement contains signal and background events described by five latent variables. Each process is a correlated Gaussian mixture with two process-specific modes and a common broad component. Independent Gaussian detector smearing produces the reconstructed inputs $\boldsymbol{x} = (x_1, \ldots, x_5)$. The common broad component guarantees overlapping analytic support and makes exact reconstructed densities available for validation. The generated samples contain 10^8 background and 2×10^7 signal events, with inclusive expected yields

$$\lambda_B^{\text{incl}} = 10^6, \qquad \lambda_S^{\text{incl}} = 1.1 \times 10^3. \tag{41}$$

A classifier preselection is trained with at most 5×10^5 events per class using three hidden layers of 256 nodes. The threshold on its estimate of p_S/p_B is selected so that the expected post-selection ratio is approximately

$B/S = 250$. The chosen cut retains 55.60% of the signal and 15.28% of the background, giving

$$\lambda_S = 611.60, \qquad \lambda_B = 152822.48. \tag{42}$$

The preselection is not part of the final likelihood; it defines the phase-space region in which all densities are normalized. Deterministic row hashes produce disjoint preselection-training, model-training, and final-evaluation partitions, and the parquet samples are streamed so the full event sets need not reside in memory.

B. Reference flow

After preselection, 5×10^5 signal and 5×10^5 background events are concatenated with equal component normalization. A normalizing flow is trained by unweighted maximum likelihood on this balanced sample. The chosen model has ten quadratic-spline coupling transforms. Each conditioner has four hidden layers with 1024 features; the splines use 16 bins over the standardized interval $[-5, 5]$ and linear tails outside it. Training uses batches of 2048, an initial learning rate of 10^{-4}, and at most 70 epochs with early stopping.

Because a smooth flow can place small probability outside the hard preselection boundary, generated events are passed through the same preselection classifier and rejected if they fail. Five million accepted events are retained as a fixed unweighted reference sample; the acceptance is 99.5548%, and its normalization is included in the accepted reference density. Against analytic balanced-mixture truth, the reference-flow log density has correlation 0.9899, root-mean-square error (RMSE) 0.2123, and mean residual -0.0122. These absolute-density residuals are precisely why the flow is treated as a proposal and corrected by discriminative ratios.

C. Process-to-reference ratios

Two classifier ensembles estimate p_S/q_ϕ and p_B/q_ϕ. Each class contains five million events: selected target MC in the numerator and flow-generated reference events in the denominator. Each ensemble has four independently initialized dense networks, each with four hidden layers of 1024 nodes, trained for at most 50 epochs with batches of 4096 and an initial learning rate of 10^{-3}. Class weights are normalized separately, and no post-training histogram calibration is applied. The arithmetic mean of the four density-ratio predictions is used in the likelihood. The final ratios are normalized on the fixed five-million-event reference sample according to Eq. (15).

Validation is performed on a common held-out partition. Score calibration tests the classifier interpretation in Eq. (6); process projections test reweighting from the reference; and reconstructed log densities are compared with analytic truth. Before empirical normalization, the mean ratios on the reference sample are 0.999149 for signal and 0.999138 for background. After normalization, the component effective sample sizes are 4.78 and 4.74 million out of five million, respectively. The hybrid log-density correlations with truth are 0.9810 for signal and 0.9848 for background, with mean residuals -0.0120 and -0.0118. The marginal reweighting closure is shown in Fig. 1; inference closure below is the more stringent test for the intended use.

D. Inference, Asimov dataset, and toys

The post-selection measurement has one POI, the signal strength μ, with intensity given by Eq. (30). A JAX-differentiable extended negative log likelihood is minimized with Minuit under $\mu \geq 0$. On the same independent, finite weighted validation measure, the hybrid and analytic reconstructed densities give $\widehat{\mu} = 1.2504$ and 1.1910, respectively. The shift attributable to the learned model on this measure is therefore 0.0594, small compared with the expected statistical uncertainty of approximately 0.56 but still visible in the profile scan in Fig. 2. Quoting both minima avoids confusing finite validation-quadrature fluctuation with neural-model bias.

Before pseudo-experiments are generated, the five-million-event reference sample is converted into the weighted Asimov dataset at $\mu_A = 1$. Its total weight is 153434.077655, agreeing with $\lambda_S + \lambda_B$ within 2.0×10^{-10}. The analytic score at the generating point is 6.8×10^{-13} and the fit returns $\widehat{\mu}_A = 1.00000000$. We obtain $q_{0,A} = 3.229623$, $Z_A = 1.7971$, and $\sigma_A = 0.556447$. A local-curvature estimate gives 0.558017, a relative difference of 0.28%.

Finally, 1000 pseudo-experiments are generated at each of $\mu = 0$ and $\mu = 1$. Independent Poisson signal and background counts are drawn, and reference indices are sampled with probabilities proportional to the normalized component ratios. At $\mu = 1$, the mean fitted signal strength is 1.0156 and its standard deviation is 0.5217; the mean observed-information estimate predicts 0.5580. The physical-boundary probability is 3.1% in toys, compared with the Asimov/Wald prediction of 3.62%. The right panels of Fig. 2 show that the Asimov prediction describes both the estimator width and the boundary-aware q_0 distribution at the precision of this toy ensemble.

E. Neural importance-sampling experiment

Exercise 6 freezes and reloads the reference flow and ratio ensembles from Exercise 5. A pilot sample of 750000 points is drawn from q_ϕ. The scan amplitude in Eq. (37) is evaluated at 13 equally spaced values over $0 \leq \mu \leq 3$, using equal a_k and dividing every Y_k by the total Asimov yield. The amplitude is floored at 10^{-3} of its median positive value and clipped at its 99.9th percentile. A sample of 500000 pilot points, resampled in propor-

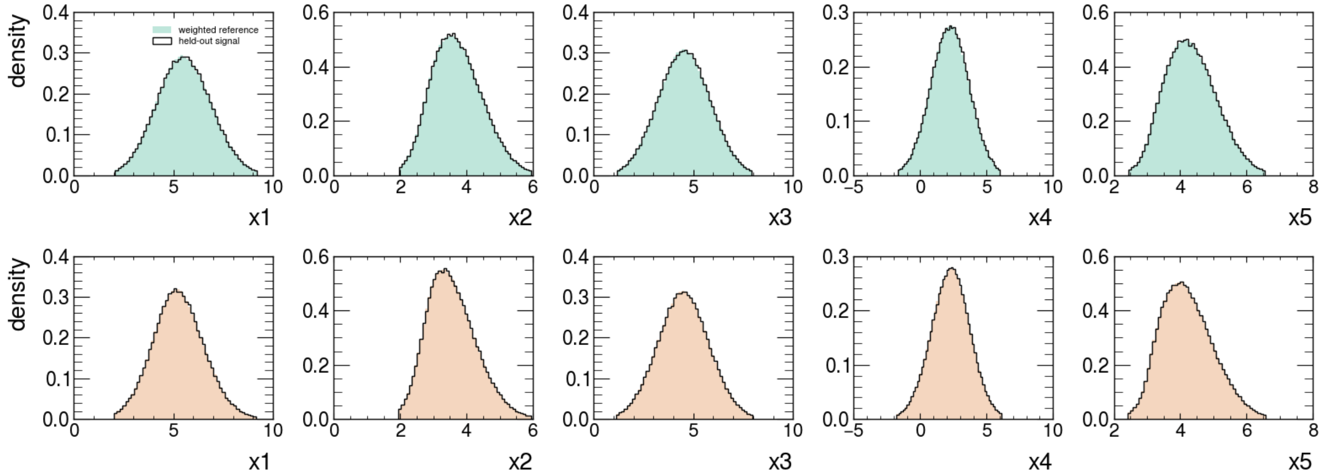

FIG. 1. Process-to-reference ratio closure after preselection. The five reconstructed features of the five-million-event flow reference are reweighted with $\widetilde{r}_S$ (top) and $\widetilde{r}_B$ (bottom) and compared with independent selected process samples. The curves are normalized densities; this test probes the learned ratios independently of the physical B/S normalization.

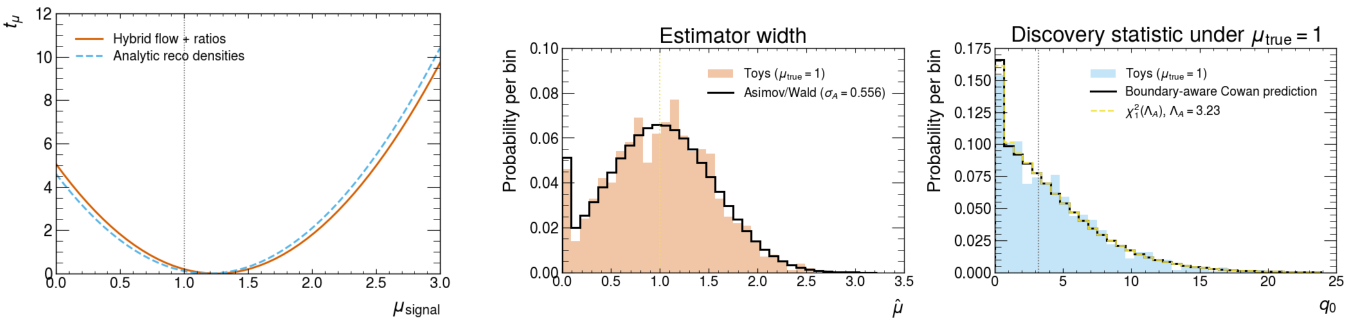

FIG. 2. Inference-level validation. Left: hybrid and analytic profile-likelihood scans on the same finite validation measure. Center: the fitted signal-strength distribution for 1000 pseudo-experiments generated at $\mu = 1$, compared with the weighted-Asimov/Wald prediction. Right: the corresponding discovery statistic compared with the boundary-aware Cowan prediction and an ordinary noncentral χ^2 distribution. The weighted Asimov fit itself closes exactly at $\mu_A = 1$.

TABLE I. Principal Exercise 5 results. Hybrid and analytic validation fits use the same finite weighted evaluation measure.

Quantity	Value
Post-selection (λ_S, λ_B)	$(611.60, 152822.48)$
Raw $(\mathbb{E}_{q_\phi}[\widehat{r}_S], \mathbb{E}_{q_\phi}[\widehat{r}_B])$	$(0.999149, 0.999138)$
Validation $(\widehat{\mu}_{\text{hybrid}}, \widehat{\mu}_{\text{truth}})$	$(1.2504, 1.1910)$
Weighted-Asimov $\widehat{\mu}_A$	1.00000000
$(q_{0,A}, Z_A)$	$(3.229623, 1.7971)$
Asimov σ_A	0.556447
Toy std$(\widehat{\mu})$ at $\mu = 1$	0.521740

tion to this amplitude, trains a smaller quadratic-spline flow with eight coupling layers, three hidden layers of 512 nodes, 16 bins, and tail bound five. The final proposal uses $\epsilon = 0.10$ in Eq. (38).

On 200000 independent reference points, the centered correlation between $\log A$ and $\log(g_\psi/q_\phi)$ is 0.645, with RMSE 0.268. This is not perfect proposal learning, so the exact defensive-mixture weights remain important. After reweighting, all five reference projections close, as shown in Fig. 3; the defensive-proposal effective sample size is $183263/200000 = 91.6\%$. The 99.9th percentile of q_ϕ/g_ϵ is 3.57, and its observed maximum is bounded by 10, as required by $\epsilon = 0.10$.

A direct-flow sample of one million events defines a high-statistics benchmark, $q_{0,A} = 3.227081 \pm 0.002568$ from ten blocks and $\sigma_A = 0.556666$. Direct and neural-importance quadratures are then compared at seven sizes from 512 to 32768, using 24 independent repetitions at every size and a 61-point scan over $0 \le \mu \le 3$. Every quadrature empirically normalizes the signal and background ratios, so the maximum absolute score at μ_A is 1.8×10^{-12} even for the smallest samples.

At the representative size $M = 2048$, the RMSE of $q_{0,A}$ decreases from 0.05954 for direct flow sampling to 0.03658 for neural importance sampling. The corresponding variance ratio is 2.69, which can be inter-

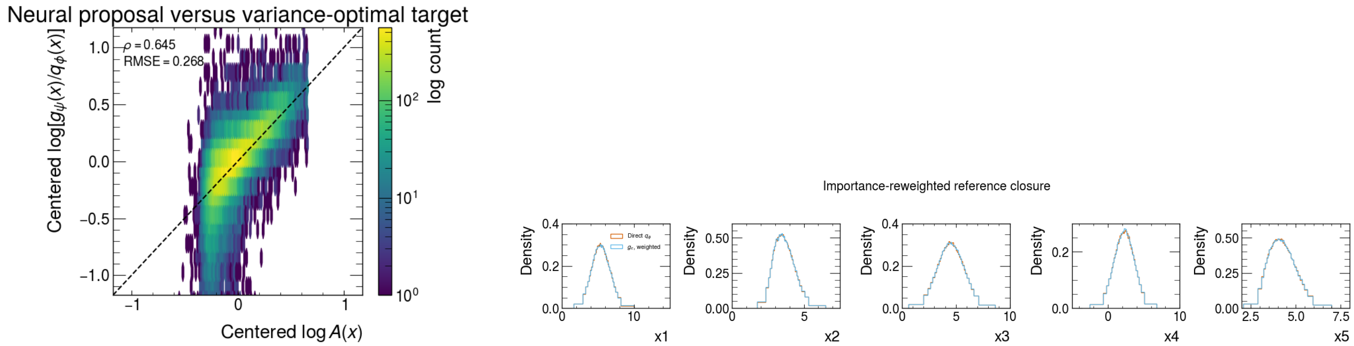

FIG. 3. Validation of the neural importance proposal. Left: the learned density displacement $\log(g_\psi/q_\phi)$ compared with the scan-amplitude target $\log A$, after subtracting their respective means. Right: direct reference-flow projections compared with samples from the defensive proposal reweighted by q_ϕ/g_ϵ. Exact reweighting restores the reference distribution despite imperfect proposal learning.

preted as the approximate direct-sampling event factor needed to reach the same precision. The RMS error over the complete likelihood scan decreases from 0.08548 to 0.05218. The gain is largest at intermediate sample sizes and approaches unity when the residual benchmark and repetition precision become limiting, as visible in Fig. 4. This experiment demonstrates variance reduction at equal numbers of likelihood-model evaluations; production studies should additionally account for proposal-training and sampling wall time.

VII. DISCUSSION AND OUTLOOK

Hybrid NSBI changes the role of absolute density estimation. Training an independent flow for every process asks each model to be accurate enough that the ratio of two imperfect densities supports a precision measurement. The hybrid approach asks one flow to cover a reference phase space and learns every precision-critical correction discriminatively. Because the denominator examples are drawn from that trained flow, classifier ratios correct its local density distortions in the population limit. At the same time, the explicit q_ϕ factor turns the ratio representation into a normalized density model.

This construction addresses the two operational limitations discussed in Section II D. The flow can generate arbitrarily many independent reference points, so numerical quadrature for profiles, impacts, and Asimov calculations is not capped by the original MC. The normalized ratios turn the same proposal into process-specific samplers, enabling pseudo-experiments and Neyman constructions without a generative model for every parameter point. Neural importance sampling adds a separate efficiency layer: it concentrates the Asimov quadrature where the expected scan receives its largest event-level contributions, while exact weights retain the original hybrid model.

Several limitations remain and should be treated as part of the method definition.

- *Support and importance-weight variance.* A flow that misses a relevant region creates extreme ratios and low effective sample size. Balanced-mixture design, tail modeling, ratio-tail monitoring, and process-wise effective sample sizes are mandatory diagnostics.

- *Classifier misspecification.* Empirical normalization fixes only the integral. Calibration, reweighting, ensemble stability, and inference-level closure against independent samples remain necessary. Neyman construction is especially valuable when residual neural bias could invalidate simple asymptotic assumptions.

- *Finite reference quadrature.* Yield normalization guarantees stationarity at the generating point but not an exact expected scan away from it. Block errors, repeated quadratures, and direct-versus-importance comparisons are needed to quantify numerical integration uncertainty separately from flow and classifier bias.

- *Negative-weight and interference components.* Individual signed contributions are not probability densities and cannot be sampled directly. A practical implementation must use positive physical combinations as sampling channels, while retaining analytic signed coefficients in the intensity, as is already done in parametrized NSBI.

- *Nuisance-parameter coverage.* A nominal reference must cover the support of all relevant systematic variations. Very large shape changes may require a broader mixture or a conditional reference flow.

The reference and integration proposals serve different objectives. Equal process normalization is a robust choice for q_ϕ because it supports classifier training and generation for every process. The neural importance flow is instead optimized for a chosen statistical task and can be retrained without changing the likelihood

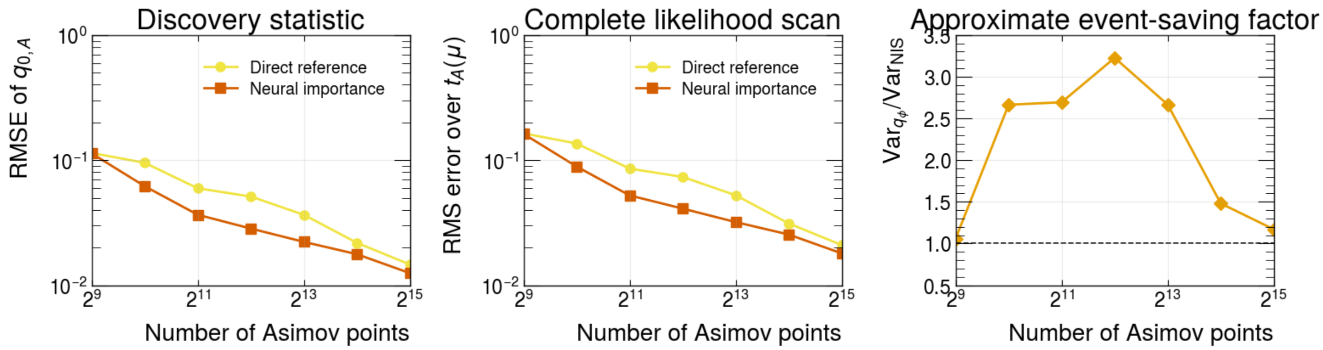

FIG. 4. Equal-size Asimov quadrature comparison over 24 repetitions. Left: RMSE of the discovery statistic relative to the million-event benchmark. Center: RMS error of the complete expected likelihood-ratio scan. Right: ratio of the direct-reference variance to the neural-importance variance. At $M = 2048$, the variance reduction is 2.69 and the complete-scan RMSE is reduced by 39%.

model. A single NIS target may cover a POI scan, several nuisance-profile trajectories, or a set of impact calculations through the weights a_k. For very high-dimensional problems, mixtures of flows, heavier-tailed bases, or iterative proposals may be preferable.

The flow map also permits randomized quasi-Monte Carlo points to be transported from the latent base distribution, which can reduce integration error when the transformed integrand is sufficiently regular [19, 20]. This strategy is complementary to neural importance sampling: the latter reshapes the sampling density, whereas quasi-Monte Carlo improves space filling within that density. Their combination is a natural next step, provided uncertainty is assessed with independent scramblings rather than with an i.i.d. standard-error formula.

VIII. CONCLUSION

We have introduced hybrid neural simulation-based inference, a construction that combines a tractable normalizing-flow reference with classifier estimates of process-to-reference density ratios. Sampling the denominator class from the learned flow makes the ratio a correction to that specific tractable proposal, so local flow error cancels in the ideal classifier limit. The resulting process densities can be evaluated, integrated, and sampled while inference retains the ratio structure of parametrized NSBI.

The hybrid representation supplies two statistical objects that are cumbersome in a ratio-only analysis: a weighted unbinned Asimov measure and a process sampler for pseudo-experiments. Matching the total Asimov weight to the expected yield supplies the extended term, and component normalization on the same quadrature guarantees exact finite-sample stationarity for yield parameters. In the demonstration, the Asimov fit closes at $\mu_A = 1$, and its $q_{0,A}$ predicts the estimator width and boundary-aware toy distribution observed in 1000 pseudo-experiments.

Finally, we formulated the expected test-statistic scan as a set of integrals and trained a second flow toward their variance-optimal proposal. At 2048 quadrature points, this neural importance sampler reduces the $q_{0,A}$ variance by a factor of 2.69 and the full-scan RMSE by 39%. The reference flow therefore makes an Asimov calculation cheap and renewable; neural importance sampling makes it efficient. Together they preserve the central advantage of parametrized NSBI—precision corrections learned as density ratios—while adding the tractable integration and generation needed for standard LHC statistical workflows.

ACKNOWLEDGMENTS

This work grew out of the simulation-based inference lectures and exercises presented at the 2026 ML4HEP School at the Tata Institute of Fundamental Research. We thank the school organizers and participants for discussions. This draft uses the public tutorial implementation in Ref. [17]. Additional acknowledgments and funding information will be added before submission.

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